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The role of primary cilia in the development and disease of the retina

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Keywords: primary cilia, ciliopathy, inherted retinal conditions, photoreceptor development, retina, intraflagellar transport

Abbreviations: ACIII, adenylyl cyclase III; ARL2, ADP-ribosylation factor-like 2 protein; ARL2BP, ADP-ribosylation factor-like 2 binding protein; ARL3, ADP-ribosylation factor-like 3 protein; ATD, asphyxiating thoracic dystrophy; BBS, Bardet-Biedl syndrome; CC, connecting cilium; CED, cranioectodermal dysplasia (Sensenbrenner syndrome); CORS, cerebellooculorenal syndrome; GC1, guanylyl cyclase 1 protein; IFT, intraflagellar transport; IS, inner segment (of a photoreceptor cell); JBTS, Joubert syndrome; LCA, Leber congenital amaurosis; MKS, Meckel-Gruber syndrome; MKKS, McKusick-Kaufman syndrome; MORM, mental retardation, obesity, retinal dystrophy and micropenis; MZSDS, Mainzer-Saldino syndrome; NINL, ninein-like protein; NPHP, nephronophthisis; OMD, occult macular dystrophy; ONL, outer nuclear layer; OS, outer segment (of a photoreceptor cell); PKD, polycystic kidney disease; ROM1, retinal outer segment membrane protein 1; RP2, Retinitis Pigmentosa 2 protein; TMEM, transmembrane protein; SLS, Senior-Løken syndrome; SNARE, soluble *N*- ethylmaleimide sensitive factor receptor; USH, Usher syndrome

The normal development and function of photoreceptors is essential for eye health and visual acuity in vertebrates. Mutations in genes encoding proteins involved in photoreceptor development and function are associated with a suite of inherited retinal dystrophies, often as part of complex multi-organ syndromic conditions. In this review, we focus on the role of the photoreceptor outer segment, a highly modified and specialized primary cilium, in retinal health and disease. We discuss the many defects in the structure and function of the photoreceptor primary cilium that can cause a class of inherited conditions known as ciliopathies, often characterized by retinal dystrophy and degeneration, and highlight the recent insights into disease mechanisms.

Introduction: The Neurosensory Retina and Photoreceptor Cells

The retina is the internal layer of the eyeball, responsible for converting light signals from the environment (focused by the anterior features of the eye) into neural impulses to be sent to the brain. This thin (0.56–0.1 mm) membrane can be further divided into two distinct layers: an inner neurosensory layer and an outer pigmented layer, the retinal pigment epithelium (RPE). The most prevalent cell-type of the retina is the neuron, of which there are three main groups responsible for relaying light generated impulses. These are the bipolar cells, the ganglion cells and the photoreceptors. Photoreceptor cells are long and narrow, and are sub-divided into inner segments (IS) and outer segments (OS) connected by a connecting cilium (CC).[1] The OS of the photoreceptor develops from a primitive primary cilium, and the OS is widely considered to be a highly modified primary cilium,[1-4] with the retinal connecting cilium homologous to the transition zone of the primary cilium.[5-7]

There are two types of photoreceptor cells, rods (**Fig. 1A**) and cones, which are named after the shape of their respective OS. These specialized regions of the cell contain high concentrations of the components of the phototransduction cascade such as the G-protein transducin and visual pigments, and low concentrations of proteins involved in other cellular functions.[8] Rod OS contain the visual pigment rhodopsin within membrane-bound discs, which stack together to form the rod shape. Cones contain several visual pigments, known as opsins, inserted into the highly invaginated plasma membranes which form the conical shape of these cells. While rod OS tips are phagocytosed daily by the RPE and new discs formed at the OS base, the cone OS are not phagocytosed in this manner. Rods are shed daily on light onset[9] while cones are shed during light offset.[10]

The CC connects the OS to the IS and consists of an axoneme of nine microtubule doublets nucleated by a triplet microtubule basal body. This is derived from the mother centriole, at the apical surface of the IS. This axoneme extends into the photoreceptor OS, converting to singlet microtubules toward the distal end (**Fig. 1A**).[11] The axoneme often reaches near the distal tip of cone OS and at least half-way along the rod OS, reaching the distal tip of rod OS in some cases.[12] The axoneme is stabilized by post-translational modifications such as glutamylation and acetylation in rods, but is turned over as membranes are replaced at the distal surface of cones.[13]

The ciliary rootlet extends from the basal side of the basal body and anchors the cilium to the cell, extending deep into IS.[14]

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It is composed of rootletin and is required for the structural stability of the photoreceptor OS (**Fig. 1A**). It is thought to also act as a docking point for transport motor proteins carrying vesicular cargo which may be destined for the CC or OS. The IS of the photoreceptors is where protein synthesis occurs, and where ATP is produced. The IS are split into two regions. The distal or ellipsoid region is packed with mitochondria, reflecting the high metabolic demands of the cells, whereas the proximal or myoid region is predominantly filled with the Golgi apparatus and the endoplasmic reticulum that enable high rates of protein synthesis in these parts of the cell. The remainder of the photoreceptors consist of the outer fiber connecting the IS to the cell body (containing the cell nucleus), which in turn is connected via the inner fiber to the synaptic terminal, forming junctions with the retinal bipolar cells for transmission of nerve impulses. These cells perform essential roles in phototransduction, and ciliary signaling is a crucial component of this activity that, when disrupted, can lead to retinal disease. The role of ciliary signaling in retinal disease has been reviewed recently[15] and the present review will instead focus on photoreceptor assembly, organization and maintenance.

Model Systems for the Study of Photoreceptor Development and Function

A range of organisms are used as models of retinal development, each with its own individual benefits and drawbacks (**Table 1**). The interested reader is referred to **Table S1** for further details. Diurnal primates are the best model of human retinal development, since they share several important features of the human retina that are absent in other mammals such as the fovea, the region of the central retina populated by cones and with the highest visual acuity.[16] Most mammals have dichroic vision, with two opsins (blue-sensitive and green-sensitive) whereas humans and diurnal primates have trichroic vision (with an additional red-sensitive opsin).[17] Ethical and practical factors prevent the use of such primates in widespread retinal research, and mice remain the most commonly-used mammalian model. However, mice do not differentiate their photoreceptor OS until P8-P16,[18] preventing the study of photoreceptor development and/or degeneration in mouse models of conditions that are embryonic or perinatal lethal, as is the case for many ciliopathies. The use of conditional knockouts by crossing to lines expressing Cre recombinase in the rods and/or cones can help to overcome such lethality. Crossing of conditional knockouts to lines expressing Cre recombinase under the control of a cone-specific gene promoter have been successfully used to develop models of foveal diseases.[19]

Additionally, mice photoreceptors lack calyceal processes (CPs), finger-like structures that protrude from the apical region of the inner segment and surround the base of the outer segment, further compromising their utility as models of retinal ciliopathies. This is particularly true in the case of mouse models of Usher syndrome (USH), an inherited condition involving sensorineural hearing loss and retinal dystrophy. Mouse knockouts of the USH1 proteins (myosin VIIa, harmonin, cadherin-23, protocadherin-15, SANS) often develop hearing loss and vestibular dysfunction but do not display retinal degeneration due to the lack of CPs.[20] Amphibians (especially *Xenopus* tadpoles) or macaques are a better model of USH1 as they possess CPs, but zebrafish are more commonly used for most models of Usher syndrome.

The retina is fully laminated and light responsive by the third day of embryogenesis in this organism, and zebrafish mutants, morphants and TALEN models provide tractable and versatile resources for the study of retinal ciliopathies. Knockouts and knock-ins by genomic editing with TALENs (or CRISPRs) are a recent exciting addition to the techniques developed for the study of photoreceptor degeneration in zebrafish. Morpholino oligonucleotide knockdown of gene transcript levels is only effective for around three days after microinjection, and so is applicable for studies of retinal development but not slower or later-onset forms of retinal degeneration. Medaka fish are also popular model organisms for studying eye development because the embryos are transparent embryos, allowing simple visual assays of mutant phenotypes, with comparatively simple and affordable animal husbandry.[21] However, whole genome duplication in fish can complicate genetic studies in this class of organisms due to the presence of multiple orthologs of genes.[22] For example, *PCDH15* (protocadherin-15) is mutated in humans with Usher syndrome type 1F[23] and both humans and other mammals have a single *PCDH15* gene that is expressed in both inner ear and retina. In contrast, the homolog has duplicated and diverged in zebrafish. The two copies of the fish gene have evolved independent, tissue-specific functions: *pcdh15a* is expressed in inner ear and mediates hearing and vestibular function, whereas *pcdh15b* is expressed in retina and mediates retinal development and function.[24] Zebrafish are therefore not an ideal model for studying this condition. Furthermore, fish and amphibian retinae also have limitations in their use for studying retinal degeneration because—in contrast to mammals—photoreceptors can regenerate in many adult fish and amphibians.[25,26] As is the case in most avenues of research, a compromise must be sought when choosing which model organism to use for retinal development and degeneration studies.

Figure 1 (See opposite page). Schematic representation of a rod photoreceptor cell and localization of ciliary proteins. (**A**) The schematic represents the rod photoreceptor cell outer segment, connecting cilium, inner segment, outer fiber, cell body, inner fiber and synaptic terminus. A number of key components of the ciliary apparatus are color coded and indicated. The IFT complex A (blue) and complex B (red) are represented in the magnified inset. A retinal pigmentary epithelial (RPE) cell is shown in gray at the top. (**B**) Confocal microscopy images of an immunofluorescent stained P20 mouse retinal cryosection showing the stratified layers of the retina. Cilium transition zone and basal body protein MKS1 is stained in green, and a novel interactant of MKS1, RNF34, is stained in red. These proteins localize to the base of the connecting cilium, as shown by the arrowheads in the enlarged insets.
(**C**) Confocal microscopy image of a human adult retinal pigment epithelium (ARPE19) cell overexpressing enhanced-GFP-tagged lebercilin and immunostained with an antibody against acetylated α tubulin, which marks the axoneme of the cilium. Lebercilin can be seen in a punctuate pattern along the axoneme. Scale bar = 10 μm.

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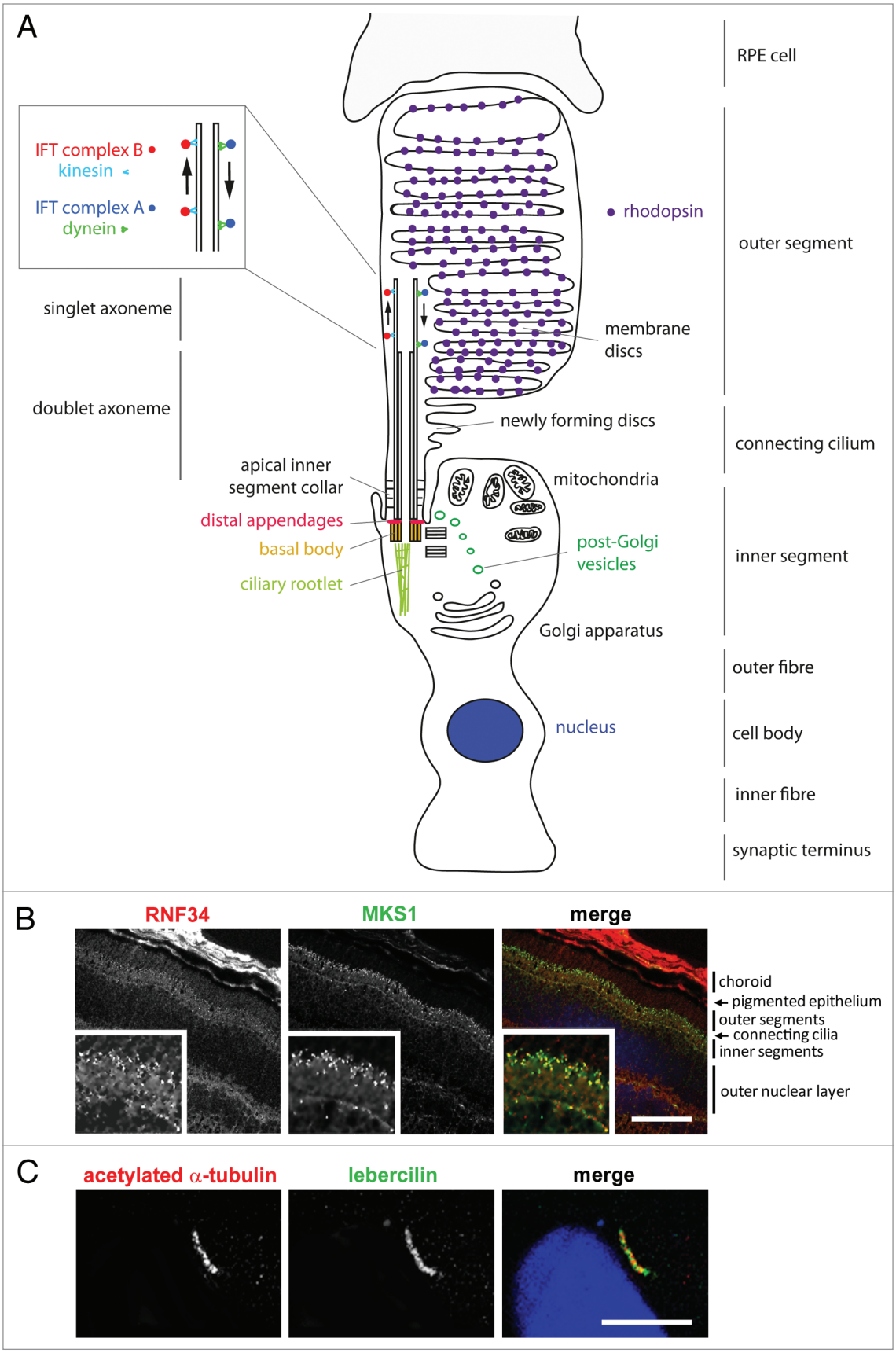


Figure 1. For figure legend see page 70.

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Photoreceptor Development and Inherited Retinal Conditions

Mutations in genes encoding proteins involved in photoreceptor development are associated with a broad group of inherited retinal dystrophies (**Table 1**), often as part of complex multi-organ syndromic conditions termed the ciliopathies.[27] Refer to **Table S1** for further details. Photoreceptor biogenesis has been suggested to occur in six distinct stages,[28] based on the four phases of ciliogenesis in an epithelial cell,[29] followed by two stages of OS development. This whole process takes over two weeks to complete in the mouse retina.[30] In the mouse, it begins with the docking of paired centrioles (each consisting of nine microtubule triplets) at the apical surface of the undifferentiated progenitor cell early in photoreceptor development.[31] The distal end of the mother centriole is surrounded by the primary vesicle and a microtubule ciliary bud begins to grow from this centriole. At the same time, this centriole matures into the basal body by recruitment of appendages, such as ODF2,[32] and aligns perpendicular to the plasma membrane. Many proteins implicated in human retinal disease are localized to the basal body (**Fig. 1B**), and may play a role in photoreceptor biogenesis. Examples include the TOPORS protein, loss of which is associated with failure of OS formation in zebrafish,[33] and retinitis pigmentosa (RP), a hereditary retinal degeneration in humans.[34] OFD1 is a distal centriole protein that regulates the length of centrioles[35] and ciliary axonemes.[36] Mutations in *OFD1* cause oro-facial-digital syndrome, non-syndromic RP, and RP as a feature of Joubert syndrome (JBTS).[37-39] *ofd1* morphant zebrafish occasionally develop retinal coloboma.[40] RAB28 localizes to the basal body and is suggested to play a role in coordinating ciliogenesis and rhodopsin transport, with mutations in this gene causing cone-rod dystrophy.[41]

Once the basal body has docked at the apical cell surface, binding of post-Golgi vesicles to the primary vesicle expands the membrane to form the ciliary vesicle, allowing the ciliary bud to extend to form the axoneme. This ciliary vesicle then fuses with the plasma membrane and the CC extends toward the RPE.[28,42] The protein CC2D2A plays a role in extension of the CC membrane through RAB8-mediated vesicular trafficking. *cc2d2a* zebrafish morphants develop disorganized photoreceptor OS and accumulation of opsins and vesicles in IS.[43] Mutations in *CC2D2A* in humans lead to RP, often as part of multi-organ syndromes.[44-46] Growth of the microtubule axoneme is dependent on chaperone proteins such as prefoldin-5 that, when mutated in mice, leads to photoreceptor degeneration.[47] Other chaperones including HSC70 (Hspa8) have also been found to be associated with photoreceptor axonemal proteins.[48] Glutamylation of axonemal tubulin is essential for normal photoreceptor development and function,[49] and the stabilization of axonemal microtubules by other post-translational modifications such as acetylation are also presumably required for these normal processes. FAM161A, a microtubule-associated protein, is thought to have a role in this stabilization process since patients with mutations in this gene develop RP.[50]

Rod OS biogenesis begins as the distal end of the CC, followed by membrane vesicular fusion to form the discs of the rod OS. Disc assembly is an on-going process as the membrane discs of the OS are constantly sloughed off to prevent the accumulation of toxic by-products of phototransduction.[51] Normal disc assembly occurs at the base of the OS, near to the ciliary axoneme. OS plasma membrane invaginates to form nascent discs, and disc proteins are then localized to their correct compartment: the OS plasma membrane, disc rim or disc lamellar region. The correct formation of discs is dependent on membrane proteins, including rom-1 and peripherin that localize to disc rims,[52,53] and the visual pigment rhodopsin.[54] RAB8 mediates transport of vesicles carrying rhodopsin to the base of the CC where they bind to the Qa-SNARE syntaxin 3, or to the IS where they bind the Qbc-SNARE SNAP-25 to fuse with the plasma membrane to deliver their cargo.[55] RP1 also plays a role in disc assembly. RP1 binds to singlet microtubules of the axoneme,[56] particularly at the point of disc membrane formation. In *Rp1* mutant mice, disc membranes are abnormally organized[57] and discs do not stack correctly. RP1 is phosphorylated by MAK, and this is thought to regulate extension of the ciliary axoneme to control CC and OS length. Loss of Mak in mice leads to increased length of photoreceptors and retinal degeneration.[58] Similar to RP1, RP1L1 binds to the photoreceptor axoneme, with consequences on OS morphology and photosensitivity, leading to progressive photoreceptor degeneration when this protein is lost in mutant mice.[59] Loss of this protein in humans is associated with occult macular dystrophy (OMD) and RP.[60,61] Myosin 7a, that is mutated in both Leber congenital amaurosis (LCA), an early-onset retinal dystrophy,[62] and Usher syndrome type 1B[63] is also localized to the site of disc morphogenesis. Myosin 7a plays a role in transport of proteins from the IS to the OS for incorporation into discs, and opsin accumulates in the IS of *Myo7a* mutant mouse.[64] The orphan membrane-bound receptor TMEM67 (meckelin), the protein product of the *TMEM67* gene, is also required for membrane disc assembly. *Bpck* mice, which have a spontaneous deletion of *Tmem67*, have structurally normal CC but dysmorphic, misaligned membrane discs, and mislocalized rhodopsin, arrestin, and transducin leading to rapid photoreceptor degeneration.[65] While the CC are structurally normal in *bpck* mice, rhodopsin transport is compromised and rhodopsin accumulates in the IS and ONL, likely reflecting a role of TMEM67 in mediating transport along the photoreceptor CC.[66] Mutations in human *TMEM67* are associated with a suite of ciliopathies, many of which include retinal degeneration phenotypes.[67-69]

Many retinal proteins also need to be correctly localized to the CC membrane. The cilium membrane is the site of many G-protein coupled receptors (GPCRs), mediating many important roles in cellular signaling. Inositol polyphosphate-5-phosphatase E (INPP5E) is a phosphatase that hydrolyses phosphotidylinositols, membrane-bound intermediate molecules for PI3K signaling. INPP5E localizes to the IS but does not appear to be required for ciliogenesis. Instead, it maintains cilium stability, and loss of the protein results in anophthalmia in mutant mice[70] and microphthalmia in morphant zebrafish.[71] Human mutations are associated with several syndromic retinal dystrophies including mental retardation, obesity, retinal dystrophy and micropenis (MORM) syndrome, JBTS and CORS.[70,72]

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Table 1. The most comprehensively characterized photoreceptor proteins mutated in human disease

Gene	Localization	Function	Retinal phenotype in animal model	Human disease (type)
AHI1	connecting cilium	thought to play a role in RAB8-mediated polarized vesicular trafficking	Ahi1 mutant mice do not develop OS, or develop abnormal OS leading to subsequent photoreceptor degeneration, associated with defects in vesicular trafficking of transducin and Rom-1, and a decrease in Rab8 expression. Opsin is also mislocalized throughout the photoreceptor is *Ahi* mutant mice. Defect in lamination of retina in zebrafish morphants.	JBTS(3)
ALMS1	basal body	thought to play a role in transport along the photoreceptor axoneme, possibly in endosome recycling	gene-trap *Alms1* mutant mice accumulate vesicles in their IS and rhodopsin is mislocalized to the OS, leading to retinal degeneration	ALMS(1), LCA
ARL6	IS, OS, ONL	regulates the BBSome	a specific isoform of BBS3 is expressed in the human retina, and when this long isoform is knocked down in zebrafish, green opsin is mislocalized, with functional consequences on vision. Bbs3L mutant mice develop disorganized IS	BBS(3), RP(55)
BBS4	IS and OPL	component of the BBSome, a regulatory submodule of IFT	*Bbbs4* null mice develop grossly normal photoreceptors at an early age but exhibit defective IFT. This leads to mislocalization of specific phototransduction proteins (rhodopsin, transducin and arrestin) but not the structural photoreceptor proteins peripherin or rom-1, which subsequently causes photoreceptor degeneration	BBS(4)
CC2D2A	connecting cilium	extension of connecting cilium membrane through Rab8-mediated vesicular trafficking	*cc2d2a* zebrafish morphants develop disorganized photoreceptor OS and accumulation of opsins and vesicles in IS. *Cc2d2a* mutant mice have microphthalmia	COACH, MKS(6), JBTS(9), RP
CEP290	base of connecting cilium	plays a role in IFT, essential for normal mislocalization of rhodopsin and arrestin but not essential for CC structure	mislocalization of rhodopsin and arrestin in *rd16* mouse. rdAc Abyssian cat. *cep290* morphant zebrafish reveal no gross lamination defects, but have statistically significant reduction in visual function.	BBS(14), LCA(10), JBTS(5), NPHP(6), MKS(4), SLS(6)
LCA5	connecting cilium	plays a role in connecting the IFT core machinery to proteins involved in selecting and recruiting cargo110	in mice lacking wild-type lebercilin, cone and rod opsins are mislocalized and the phototransduction G-protein transducin partially mislocalizes in response to light.	LCA(5)
MAK	base of connecting cilium	kinase, phosphorylates RP1 to regulate extension of ciliary axoneme to control CC and OS length	loss of Mak in mice leads to increased length of photoreceptors and retinal degeneration	RP(62)
MKKS	connecting cilium	chaperonin-like BBS protein, thought to regulate BBSome assembly	late-onset photoreceptor degeneration is observed in Bbs6 mutant mice	MKKS, BBS(6)
MYO7A	base of connecting cilium, at the site of disc morphogenesis	plays a role in transport of proteins from the IS to the OS for incorporation into discs	opsin accumulates in the IS of *Myo7a* mutant mouse	USH(1B), LCA
NPHP1	connecting cilium, especially around the basal body	thought to play role in the transport of specific proteins along the photoreceptor	*Nphp1* mutant mice exhibit defects in sorting of proteins between the photoreceptor IS and OS, and compromised IFT	NPHP(1), SLS(1), JBTS(4)
NPHP4	connecting cilium. Proximal to RPGRIP1, RPGR and SDCCAG8	interacts with CEP290 and RPGRIP1.	*Nphp4* mutant mice mislocalize rhodopsin and ROM-1 to the IS and INL, do not develop normal OS despite developing normal CC. The photoreceptors degenerate rapidly. Synaptic ribbons develop normally but degenerate. Mislocalization of synaptic vesicle protein and post-synaptic density protein. CORD in Nphp4 wire-haired daschunds.	NPHP(4), SLS(4), CORS

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Table 1. The most comprehensively characterized photoreceptor proteins mutated in human disease (continued)

Gene	Localization	Function	Retinal phenotype in animal model	Human disease (type)
TMEM67	exact localization in photoreceptors not known	required for membrane disc assembly and rhodopsin transport	dysmorphic, misaligned membrane discs, and mislocalized rhodopsin, arrestin, and transducin, leading to rapid photoreceptor degeneration in *bpck* mice. CC structurally normal but rhodopsin mislocalized to IS and ONL in bpck mice.	COACH, JBTS(6), MKS(3), NPHP(11)
TOPORS	basal body	E3 ubiquitin ligase function. Role in photoreceptors unclear	morpholino silencing in zebrafish affects retinal development, OS fail to form	RP(31)
TTC8	connecting cilium	component of the BBSome, a regulatory submodule of IFT	retinal degeneration and mislocalization of rhodopsin to IS in *Bbs8* mutant mice	BBS(8), RP(51)
RP1	photoreceptor axoneme	binds to singlet microtubules of the axoneme, especially at the point of disc membrane formation	In *RP1* mutant mice, disc membranes are abnormally organized and discs do not stack correctly.	RP(1)
RP1L1	photoreceptor axoneme	interacts with RP1, thought to play similar role to RP1	abnormal OS morphology and photosensitivity in *RP1L1* mice	OCMD
RP2	basal body	acts as the ARL3 GTPase activating protein. Plays a role in trafficking of vesicles from Golgi to cilium	small eyes and retinal degeneration in *rp2* zebrafish morphants	RP(2)
RPGR	connecting cilium	contains a domain with homology to the RCC1 guanine nucleotide exchange factor (GEF) for Ran GTPase. This RCC1-like domain in RPGR acts as a GTP/GDP exchange factor for RAB8, a GTPase important for vesicular trafficking to the primary cilium	mislocalization of cone opsins in the cell body and synapses; reduced levels of rhodopsin in rods; leading to photoreceptor degeneration in *Rpgr* knockout mice	RP(3), CORD(X1)
RPGRIP1	distal part of the photoreceptor axoneme in rod and cone OS in humans, CC in mice.	transprt along the photoreceptor. Recruitment of RPGR, NPHP4 and SDCCAG8 to the CC	mice lacking RPGRIP1 develop grossly oversized OS discs. *Rpgrip1nmf247* mice lack RPGR, NPHP4 and SDCCAG8 at the CC.	LCA(6), CORD(13)
TTC21B	photoreceptor axoneme	intraflagellar transport A complex protein	*Ttc21b* mutant mice have defective photoreceptor development	ATD(4), NPHP(12)
USH1B (myosin7a)	site of disc morphogenesis. Ribbon synapse. RPE. Apical IS collar.	molecular motor. plays a role in transport of proteins from the IS to the OS for incorporation into discs and RPE65 transport	opsin accumulates in the IS of *Shaker1* (*Myo7a* mutant) mouse. *Myo7a* mutant mice have lower levels of RPE65, the RPE isomerase that has a key role in the retinoid cycle. Eyes not studied in *mariner* zebrafish mutant.	USH(1B)
USH1C (harmonin)	at the apical IS collar and ribbon synapse	structural protein, functions in the docking and loading of IFT cargos	*deaf circular* (*Dfcr*/*Ush1c*) mice do not undergo retinal degeneration and have normal synaptic ultrastructure and ERGs. Mice with knock-in of c.216G > A cryptic splice site mutation in Exon 3 of *Ush1c* have progressive loss of rods between 6 and 12 mo of age	USH(1C)
USH2A (usherin)	at the apical IS collar, basal body of connecting cilium	structural protein, functions in the docking and loading of IFT cargos, essential for long-term structural maintenance of photoreceptors. Interacts with lebercilin and ninein-like protein	mice lacking Ush2a develop late-onset progressive photoreceptor degeneration	USH(2A)
USH2C (GPR98, VLGR1)		G-protein coupled receptor. Exact function in photoreceptors not known	mild, late onset abnormalities in retinal function in *Vlgr1/del7TM* mice	USH(2C)

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Trafficking of INPP5E to the cilium depends on PDE6D, in concert with ARL13B and CEP164,[73] which when mutated in humans also lead to syndromic retinal dystrophies.[74,75]

The Tectonic proteins TCTN1 and TCTN2 play a role in targeting GPCRs to the cilium membrane, as well as other components of G-protein signaling such as the downstream effector adenylyl cyclase III (ACIII) and the putative channel protein polycystin-2[76]. *Tctn1* mutant mice develop microphthalmia and mutations in humans lead to JBTS.[76] *Tctn2* mutant mice also develop micropthalmia, and mutations in humans are associated with JBTS[77] and Meckel-Gruber syndrome (MKS).[78]

Transport in the Photoreceptor

All protein synthesis in the photoreceptor occurs in the IS, yet the OS is the region involved in photoreception, with the highest demand for protein. The mouse OS proteome contains almost 2000 proteins,[79] yet transcription and translation occur in neither the OS nor CC. An estimated 10% of protein is lost from the rod OS each day as membranes are shed into the RPE to prevent accumulation of toxic by-products of phototransduction.[30,80] Thus, transport between the two segments along the CC is of critical importance to photoreceptor development and function.[51,81]

Some molecules, including even some proteins, move along the CC by passive diffusion. This is perhaps surprising given the presence of the septin "barrier" at the base of the cilium that is thought to regulate the movement of molecules along the cilium.[82] However, it has been shown that the CC does not significantly inhibit protein diffusion through the photoreceptor cell.[83] There remains a debate over whether transducin and arrestin, proteins involved in phototransduction, move between the OS and IS along the CC in response to light by either active or passive transport mechanisms.[84-87] In at least some cases, proteins move by diffusion. For example, after light-induced translocation, transducin forms a stable complex with UNC119 and diffuses back into the OS.[88] Movement of transducin is also thought to be regulated by the Ca^{2+}-binding centrin proteins that are localized to the basal body and axoneme of the CC[89] and are phosphorylated by CK2.[90] UNC119 also binds myristoylated ciliary proteins, such as nephrocystin-3 (NPHP3), to target them to the ciliary membrane and maintain the spatial organization of the cilium. UNC119 releases its protein cargo when it binds to the activated isoform of the small GTP-binding protein ARL3-GTP, but not when it binds ARL2. The Retinitis Pigmentosa 2 protein (RP2) localizes to the Golgi apparatus and the basal body of the photoreceptor, and coordinates vesicle trafficking and docking at the base of the cilium.[91] RP2 acts as the ARL3 GTPase activating protein (GAP)[92] and ARL2BP is an effector for both ARL2 and ARL3. Correct targeting of such proteins to ciliary membranes is functionally important for photoreceptors, and mutations in RP2, NPHP3 and ARL2BP can all lead to ciliopathies involving retinal degeneration in humans and model organisms.[93-97]

Intraflagellar Transport in the Photoreceptor Axoneme and Inherited Retinal Conditions

ARL3 is also involved in the main active transport process along the photoreceptor axoneme: the process of intraflagellar transport (IFT).[98,99] IFT is required for the entire process of CC and OS development, and IFT proteins are involved at all stages of this development. A pool of IFT proteins is localized in the cytoplasm at the base of the axoneme during the formation of the ciliary vesicle and elongation of the axoneme early in CC and OS development, and IFT proteins are consistently associated with the developed axoneme.[28] IFT is also essential for normal photoreceptor function by mediating the transport of proteins of the phototransduction cascade to the OS.[87] Cargoes of IFT have been shown to include rhodopsin, chaperone proteins and the photoreceptor-specific membrane protein guanylyl cyclase 1 (GC1, Gucy2e).[48]

IFT is a bi-directional transport process that moves cargo along the axoneme of the cilium from base to tip (anterograde IFT, from IS to OS in the case of the photoreceptor), and from tip to base (retrograde IFT, from OS to IS). It is likely that retrograde IFT plays a less important role than anterograde IFT in photoreceptor function, because proteins are shed from the distal tip of the OS rather than transported back to the IS for recycling. Nevertheless, retrograde IFT components are found to be associated with the CC, as the cell will always have a minimum requirement for retrograde IFT to return anterograde IFT proteins back to the IS. Kinesins catalyze anterograde IFT,[100,101] whereas dyneins (specifically the axonemal dyneins and the DHC1b isoform of cytoplasmic dynein) power retrograde IFT.[102] The retrograde IFT motor component cytoplasmic dynein 2 is required for OS biogenesis and function, and is found localized to the photoreceptor axoneme.[103] Zebrafish lacking this cytoplasmic dynein develop short, disordered OS which accumulate vesicles.[104]

Heterotrimeric kinesin-II, composed of subunits of KIF3A in addition to either KIF3B or KIF3C, has been shown to be particularly important for IFT and is considered the most important kinesin motor for powering anterograde IFT. KIF3A localizes to the axoneme of the CC and IS in fish,[105] *Xenopus*, monkey, and humans[106] and KIF3B localizes to the CC and IS in *Xenopus* retina.[106] Knockout of *Kif3a* or *Kif3b* in mouse results in mid-gestation embryonic lethality, reflecting the general importance of these motor subunits in anterograde IFT in all cilia.[107,108] Photoreceptor defects are not seen in *Kif3a* or *Kif3b* heterozygote mice, suggesting that variants in these genes in humans do not contribute to degenerative human retinal conditions.[109] However, mice with a conditional knockout of *Kif3a* in rods and cones accumulate vesicles and opsin in the IS preceding photoreceptor death.[87] The accumulation of opsin is less severe in the rods yet cell death is more rapid in these cells, suggesting that kinesin-II plays different roles in opsin transport in rods and cones.[110] *Kif3c* knockout mice, however, are viable[111] and do not exhibit photoreceptor defects,[109] suggesting redundancy with either Kif3a or Kif3b. Consistent with this, *kif3b* morphant zebrafish embryos develop relatively short but otherwise functionally normal

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photoreceptor cilia, also suggesting some sort of functional redundancy. Overexpression of *kif3c* in these *kif3b* morphants confirmed partial functional redundancy of these subunits.[112] Homodimeric kinesin-II, composed of Kif17, is also thought to play a role in anterograde IFT, although its function varies in different cilia. Kif17 is thought to play a role in photoreceptors, as it is found in the axoneme of mouse photoreceptor cells and expression of a dominant negative form of *kif17* in zebrafish affects the development of photoreceptor OS.[113] Similarly, *kif17* morphant zebrafish develop short OS and aberrant discs.[114]

In association with the motor proteins, two distinct IFT complexes are associated with the bidirectional transport (**Fig. 1A**). Proteins involved in anterograde IFT are collectively known as IFT complex B proteins, and those involved in retrograde IFT are components of the IFT complex A. IFT complex B proteins IFT20, 52, 57 and 88 localize around the basal body and in discrete puncta along the axoneme of the CC of retinal photoreceptors.[115,116] All of these IFT components are required for normal OS development and retinal function. Loss of *ift52*, *57,* or *88* in zebrafish leads to an absence of OS and subsequent photoreceptor degeneration.[117] IFT57 is not essential for IFT, but is required for efficient IFT as it plays a specific role in dissociation of kinesin-II from IFT complex proteins.[118] IFT57 interacts with DYF-1/Fleer,[119] as does IFT74, which helps the IFT particle dock onto the anterograde kinesin motor.[120] *Ift88* hypomorphic mutant mice develop rod cell photoreceptors with abnormal OS and mislocalized opsin, leading to progressive retinal degeneration.[115] Mutations in IFT80, another IFT complex B protein, cause the skeletal ciliopathy asphyxiating thoracic dystrophy (ATD, also known as Jeune syndrome) in humans, a condition that involves retinal degeneration.[121] Zebrafish *ift80* morphants exhibit defects in photoreceptor OS formation and photoreceptor death[122] but hypomorphic *Ift80* mutations in mice (human mutations associated with ATD are hypomorphic) have normal retinas at P21, although retinal degeneration may occur later.[123]

IFT20 is an IFTB component that has both distinct and complementary functions to other IFT proteins since it appears to shuttle between the basal body and the Golgi apparatus,[124] yet it is also essential for normal photoreceptor development and function. IFT20 localizes to the cytoplasm around the basal body of the CC of the retinal photoreceptors[125] and *Ift20* knockout mice do not traffic opsin appropriately and OS develop abnormally.[126] Similarly, the IFTB component Ttc26 is required for OS formation, and *ttc26* zebrafish morphants have short or absent OS.[127] The exact role of IFTB component TRAF3IP1 (IFT54) and IFT172 in photoreceptor development is unclear, but *Traf3ip1* mutant mice develop small eyes and *Ift172* mutants do not develop eyes at all, suggesting the critical importance of these proteins in ocular development.[128,129] It is likely that the other IFTB components are also required for photoreceptor development, but to date their roles in this process have not been studied, in some cases because loss of the protein is embryonically lethal in mice.[130]

IFT complex A protein IFT140 localizes along the photoreceptor axoneme and is particularly abundant at the tip of the axoneme.[125] IFT140 is essential for normal photoreceptor development in humans. Mutations in human *IFT140* are a cause of Mainzer-Saldino syndrome (MZSDS)[131] and ATD,[132] both of which are skeletal ciliopathies that are associated with retinal dystrophy phenotypes. TTC21B, another IFT complex A protein, also localizes to the photoreceptor axoneme, and loss of this protein in mutant mice leads to defective photoreceptor development.[133] Mutations in human TTC21B are also associated with ATD, involving photoreceptor degeneration.[134] WDR19, which links IFTA to the BBSome (see below), is proposed to be a regulatory sub-module of IFT.[135] WDR19 is also mutated in ATD and cranioectodermal dysplasia (CED, also known as Sensenbrenner syndrome), which occasionally involves photoreceptor dystrophy.[136] It remains unknown why a skeletal ciliopathy, namely ATD, can be caused by recessive mutations in both an IFTB protein (IFT80) and IFTA proteins, and why they predominately affect the endochondral ossification of the long bones.

Other proteins external to the core A and B complexes regulate IFT, including components of the BBSome that are also localized to the base of the photoreceptor axoneme. The BBSome, comprised of BBS1, 2, 4, 5, 7, 9, and TTC8 (BBS8), is thought to function as a coat complex to sort cilium proteins into membrane domains.[137] Another class of BBS proteins, the chaperonin-like proteins MKKS (BBS6), BBS10 and BBS12, are thought to regulate the assembly of the BBSome[138] and LZTFL1 (BBS17) regulates trafficking of the BBSome into the cilium.[139] The activity of the BBSome is regulated by Arf-like GTPase ARL6 (BBS3)[137] which is found throughout the IS, OS and ONL.[140] Many BBS proteins have been shown to localize to the photoreceptors,[140-143] and loss of most of the BBSome proteins leads to retinal degeneration in model organisms and humans as part of the multi-organ condition Bardet-Biedl syndrome (BBS).[141,144-159] For example, *Bbs1* knock-in mutant mice develop disorganized OS, and exhibit degeneration of IS and OS.[160] Other work has exemplified the disease mechanism of disrupted trafficking that underlies the retinal degeneration in this ciliopathy. *Bbbs4* null mice develop grossly normal photoreceptors at an early age but exhibit defective IFT. This leads to mislocalization of specific phototransduction proteins (rhodopsin, transducin and arrestin), but not the structural photoreceptor proteins peripherin or ROM1, which subsequently causes photoreceptor degeneration.[142] *Bbs2* mutant mice similarly mislocalize rhodopsin, leading to retinal degeneration.[161] Late-onset photoreceptor degeneration is observed in *Bbs6* mutant mice[162] and *Bbs8* mutant mice also mislocalize rhodopsin to the IS, leading to retinal degeneration.[163] BBS7 is a core component of the BBSome and physically interacts with the BBS chaperonin complex, and *Bbs7* mutant mice exhibit IS, OS and ONL degeneration[164] due to defects in membrane protein trafficking. However, in contrast, *bbs9* morphant zebrafish have an even more severe phenotype and the retina does not develop OS or discrete layers.[165]

Proteins Interacting with IFT Components in Photoreceptors

Many other proteins, while not core components of IFT particles, are associated with IFT complexes. These proteins include

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the Arf-like GTPase ARL13, which is thought to promote association between IFT complex A and complex B.[98] Arl13B localizes to the CC in mice[74] and *Arl13b*^{hnn} mutant mice develop abnormal eyes, polydactyly and neural tube patterning defects.[166] Mutations in *ARL13B* are a cause of the severe neurodevelopmental condition Joubert syndrome (JBTS) that can include additional clinical features such as RP.[74] Lebercilin also interacts with complex A, complex B IFT proteins and retrograde IFT motor proteins, and is found to localize along the ciliary axoneme (**Fig. 1C**). In photoreceptor cells it is localized at the base of the CC, where it is thought to regulate loading and unloading of IFT cargo. This hypothesis is supported by the finding that in mice lacking wild-type lebercilin, cone and rod opsins are mislocalized and transducin, the G protein mediating phototransduction, partially mislocalizes in response to light. This suggests that lebercilin plays a role in connecting the IFT core machinery to proteins involved in selecting and recruiting cargo.[167] Lebercilin is not considered a core component of the IFT machinery because siRNA knockdown of *LCA5*, the gene which encodes lebercilin in humans, does not affect ciliogenesis or IFT88 localization in ciliated retinal cell lines. These interactions are disrupted when the protein is truncated, leading to disrupted IFT. This results in Leber congenital amaurosis (LCA), an early-onset retinal dystrophy in humans.[167,168] In at least some patients with mutations in this gene, photoreceptors in the central retina are structurally normal[169] but visual acuity is poor from an early age. This suggests that, while important for IFT along the photoreceptor axoneme, lebercilin is not essential for structural development of the photoreceptor.

Lebercilin interacts with ninein-like protein (NINL) isoform b and usherin (USH2A) isoform b at the basal body of the connecting cilium, where these proteins are thought to play a role in regulating the docking of IFT particles.[170] NINL isoform b is a centrosomal protein that functions in nucleation, anchoring and outgrowth of microtubules,[170] and usherin (encoded by *USH2A*) is essential for the long-term structural maintenance of photoreceptors. Mutations in *USH2A* cause Usher syndrome[171] and mice lacking *Ush2a* develop late-onset progressive photoreceptor degeneration.[172] The usherin protein localizes to the apical IS membrane that wraps around the CC, a specialized region also known as the apical inner segment collar or pericilium that is homologous to the periciliary ridge complex of amphibians. Myosin VIIa (USH1B) also localizes to this apical IS collar,[173] as does VLGR1 (USH2C), whereas the USH proteins harmonin (USH1C), SANS (USH1G), and whirlin (USH2D) act as structural proteins at the apical IS collar. The apical IS collar is a specialized membrane domain that functions in the docking and loading of IFT cargos, and USH proteins are therefore considered to play important roles at this stage of IFT.[174] Their loss or mutation results in deafness and photoreceptor degeneration in humans.[63,175-178]

CEP290 is also believed to play a role in IFT along the CC because truncating mutations in *CEP290* lead to mislocalization of rhodopsin and arrestin.[179] CEP290 is necessary for transport along the CC, but is not required for CC development as loss of CEP290 does not compromise normal CC structure.[179] CEP290 is localized to the basal body, pericentriolar matrix and axoneme of the CC. Mutations in *CEP290* are associated with a suite of ciliopathies with retinal involvement, ranging from LCA[180] to multi-organ JBTS.[181,182] Mutations in this gene are the single most common cause of non-syndromic LCA, accounting for 21% of cases[180,183] *CEP290*-mutated LCA patients retain normal overall retinal and photoreceptor architecture in the cone-rich central retina, but rod-rich pericentrical and peripheral regions are not conserved in this manner. CEP290 interacts with RAF-1 kinase inhibitory protein (RKIP), and accumulation of Rkip in photoreceptors of *Cep290* mutant mice is thought to be one of the causes of photoreceptor degeneration in these mice.[184]

CEP290 also interacts with RPGR, another protein that localizes to the CC[185] and is thought to interact with several protein complexes that mediate transport along the photoreceptor. CEP290 also interacts with the USH protein complex through whirlin,[186] and multiple components of the NPHP complex, possibly as two distinct complexes involving NPHP1, 2, 5 and NPHP4, 6, 8.[187] However, most of the insights about potential disease mechanisms have come from further study of the complexes containing RPGR. *RPGR* is mutated in 15% of RP cases in humans[188] and loss of this protein in mice leads to ectopic localization of cone opsins and the reduction in levels of rhodopsin in rods, leading to cone-rod degeneration.[189] RPGR contains a domain with homology to the RCC1 guanine nucleotide exchange factor (GEF) for Ran GTPase. This RCC1-like domain in RPGR acts as a GTP/GDP exchange factor for RAB8, a GTPase important for vesicular trafficking to the primary cilium.[190] Jouberin, the protein encoded by *AHI1* and mutated in JBTS,[191] localizes to the CC and is thought to play a role in this RAB8-mediated polarized vesicular trafficking. *Ahi1* mutant mice either do not develop OS or develop abnormal OS leading to subsequent photoreceptor degeneration, associated with defects in vesicular trafficking of transducin, ROM1 and opsin with a decrease in *Rab8* expression.[192,193] *ahi1* morphant zebrafish develop abnormally shaped eyes with coloboma and defective retinal lamination.[194]

CEP290 and RPGR also interact with RPGR-interacting protein 1 (RPGRIP1) in rod OS.[195] Loss of RPGRIP1 has no effect on central (cone-rich) retinal architecture[196] but it is required for rod OS structure[197] and mice lacking RPGRIP1 develop grossly oversized OS discs.[198] RPGRIP1 is localized to the distal part of the photoreceptor axoneme, and is required for the recruitment of RPGR, NPHP4 and SDCCAG8 to the CC: *Rpgrip1*^{nmf247} mutant mice lack these proteins at the CC.[199,200] Mutations in *RPGRIP1* in humans are associated with RP,[201] cone-rod dystrophy (CORD)[202] and LCA.[203] NPHP4 mutations lead to nephronophthisis (NPHP) and RP,[204] and SDCCAG8 is mutated in BBS and a retinal-renal ciliopathy.[205-207] Mutations in the homolog of *RPGRIP*, *RPGRIP1L*, are also associated with a suite of disorders involving retinal dystrophy, including JBTS,[208] MKS and cerebello-oculo-renal syndrome (CORS).[209] A specific mutation in *RPGRIP1L* has been shown to be a modifier of retinal phenotype in these syndromic ciliopathies.[210] Mutation of alanine 229 to threonine compromises the interaction of RPGRIP1L with RPGR, with functional consequences on transport within the

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photoreceptors that leads to retinal degeneration. RPGRIP1L also interacts with MKS and NPHP proteins at the ciliary transition zone/basal body to initiate IFT-driven cilium extension early in ciliogenesis.[211] The *Ftm* mouse mutant (in which *Rpgrip1l* is mutated) develops microphthalmia, associated with a global reduction in cilia number and Shh signaling defects.[212]

ALMS1, which is mutated in Alström syndrome (ALMS), a condition often presenting with CORD[213,214] and LCA,[62] is thought to play a role in transport along the photoreceptor axoneme, as gene-trap *Alms1* mutant mice accumulate vesicles in their IS and rhodopsin is mislocalized to the outer nuclear layer.[215] ALMS1 has been shown to play a role in endosome recycling in other cell types, which may reflect its role in photoreceptors.[216] B9D2, inversin (NPHP2) and IQCB1 (NPHP5) are also thought to play roles in the transport of specific proteins along the photoreceptor. IQCB1 (NPHP5) localizes to the CC, OS and basal bodies of human and mouse photoreceptors, it interacts with RPGR in the CC,[217] and mutations in *NPHP5* are the most common cause of Senior-Løken syndrome (SLS), a ciliopathy that associates NPHP with RP or LCA.[217] Only about 10% of individuals with NPHP also have RP or LCA, constituting SLS, but 100% of NPHP patients with *NPHP5* mutations have RP. This emphasizes the essential role of IQCB1 in the photoreceptor,[217] but, by contrast, RP is very rarely seen in NPHP2 patients, suggesting that inversin has a high level of functional redundancy in human photoreceptors.[218,219] Loss of *b9d2* or *nphp2* (inversin) in zebrafish leads to mislocalized opsin but not mislocalized peripherin,[119] whereas *B9D2* mutations in humans are a cause of the lethal ciliopathy MKS that can occasionally present with retinal involvement.[220] Nephrocystin (NPHP1) localizes to the CC, especially around the basal body, and may play a similar role to B9D2 and inversin.[6] Indeed, *Nphp1* mutant mice exhibit defects in the both the sorting of proteins between the photoreceptor IS and OS, and compromised IFT.[221] Mutations in *NPHP1* cause the ciliopathies NPHP,[222] JBTS[223] and SLS,[224] all of which include retinal degeneration in some patients. NPHP4 localizes to the base of the connecting cilium, proximal to SDCCAG8, RPGRIP1 and RPGR[200] and when mutated leads to CORD in dogs,[225] NPHP, SLS and CORS in humans.[204,226,227] Similar to *Nphp1* mutant mice, mice with mutations in *Nphp4* exhibit mislocalization of OS proteins (rhodopsin and ROM1) to the IS and inner nuclear layer. These mice do not develop normal OS despite developing normal CC, and their photoreceptors degenerate rapidly leading to early-onset loss of vision.

Interestingly, the synaptic ribbons of *Nphp4* mutant mice also degenerate after developing normally, with associated disturbances in the localizations of synaptophysin (a major synaptic vesicle protein) and post-synaptic density protein 95 (PSD95, a presynaptic marker in the outer plexiform layer).[228] KIF3A and IFT88 are also localized to the ribbon synapse, and expression of dominant-negative KIF3B results in a complete loss of ribbon synapses,[113] suggesting that IFT has an essential functional role at the photoreceptor synapse.[115,229] The presence of clarin-1 (USH3A), which is mutated in Usher syndrome, further implicates the role of cilia proteins at the ribbon synapse in inherited disease.[230]

Future Perspectives and Challenges

Further extending the role of IFT and the cilium in the retina, Ataxia-telangiectasia and Rad3 (ATR) protein has recently been shown to be localized to the photoreceptor CC, and mice with mutations in this gene exhibit early-onset degeneration of rods and cones.[231] ATR is a DNA damage sensor, potentially extending the role of photoreceptors, and primary cilia in general, to involvement in the DNA damage response (DDR). Supporting this hypothesis, mutations in other DDR genes (*CEP164* and *ZNF423*) have been found in ciliopathy patients with retinal degeneration. It has been suggested that defective CC could cause postnatal accumulation of UV light-induced DNA damage, subsequently leading to retinal degeneration.[75] However, the details of this novel ciliopathy disease mechanism remain unclear, and will undoubtedly form the basis of exciting advances in basic research.

These recent findings highlight our ever-expanding understanding of the importance of primary cilia in retinal development and function, and their role in human disease (**Table 1**). Refer to **Table S1** for further details. Novel uncharacterised proteins are still being discovered to play a role in such processes, such as C8orf37, a ciliary protein which is mutated in CORD and RP[232] and C2orf71, a ciliary protein mutated in RP in humans and progressive retinal atrophy in several breeds of dog.[233,234] While our understanding of primary cilia and specialized sensory cilia such as the photoreceptor has advanced greatly in the past two decades, there still remains much to discover. With the ever-increasing power and affordability of genetic sequencing technologies, there are now unprecedented opportunities for rapid gene discovery in this group of retinal conditions, providing further insights into disease mechanism. A better understanding of the genetic basis of eye disease provides opportunities for improved diagnostics and disease management, including gene therapy. Retinal disease is a convenient model for the study of gene therapy, since vectors carrying wild-type copies of mutated genes are easily delivered to the affected cells by subretinal injection, and this method has been used to successfully restore partial vision in patients with a variety of gene mutations. Such therapies are currently in development for the treatment of patients with *LCA5* mutations (Ronald Roepman, personal communication) because in this ciliopathy, as in many retinal ciliopathies, retinal architecture is not compromised and blindness arises due to the lack of a single specific protein causing defects in protein trafficking. Preliminary gene therapy experiments in *Bbs1* mice have given promising results, with improved rhodopsin trafficking, reduced photoreceptor degeneration and improved ERG.[235] Similarly, *Bbs4* mutant mice have been successfully treated with gene therapy to restore rhodopsin localization, OS structure and prevention of photoreceptor degeneration.[236] While treating BBS has its difficulties (particularly in ensuring that the correct stoichiometry of the BBSome complex is restored), retinal gene therapy remains a

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promising therapeutic approach that needs further research to assess its clinical efficacy and utility.

However, gene therapy is costly and difficult to develop, and can only be used to treat individuals with specific mutations in specific genes. This is a considerable disadvantage for groups of conditions with broad genetic heterogeneity such as the ciliopathies. The wide phenotypic variability, including allelism in different clinical entities, and the genetic heterogeneity of these conditions remain major challenges in the field of research into photoreceptor development and disease, especially in the attempts to develop therapies for these conditions. Broader therapeutic interventions such as treatment with tauroursodeoycholic acid (TUDCA), an anti-apoptotic factor, may offer opportunities for all degenerative retinal ciliopathies in the medium term, and it is promising that this treatment has been used successfully to slow retinal degeneration in *Bbs1* mice.[237]

Summary

The photoreceptor OS is a highly modified primary cilium which has evolved to play a crucial role in vision. Development of the photoreceptor is a complex process that is an elaborate form of the ciliogenesis seen in other cell types. Most of the transport along the photoreceptor is performed by the cilium transport process of intraflagellar transport (IFT). IFT is essential for the movement of proteins involved in visual transduction in this cell type. For these reasons, proteins involved in primary cilium growth, structure, maintenance and function are of critical importance to retinal structure and function. The loss or mutation of these proteins results in a group of conditions known as the retinal ciliopathies that have defects in retinal development or retinal degeneration. Many of these proteins also have functions in the primary cilia of other cell types, so mutations are often associated with other complex multi-organ or developmental phenotypes, and retinal dystrophy forms just part of a syndromic ciliopathy. A range of model organisms and human genetic studies have advanced understanding of the role of the photoreceptor cilium in retinal health and disease, but a number of key challenges still remain in this field of research. Further insights into disease mechanisms in this group of conditions, as well as the identification of mutations in an ever-increasing number of genes, will be essential if gene therapy is to fulfil the early promise of many research studies and become an effective clinical treatment for the retinal ciliopathies.

Disclosure of Potential Conflicts of Interest

No potential conflicts of interest were disclosed.

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Supplemental Materials

Supplemental materials may be found here:
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